

HDOC DAY-10 ACTIVITY REPORT

BATCHES 15 & 16

PART 1 Time Converter Program	PART 2 Leap Year Checker	PART 3 Compilation & Execution
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Organized submission copy with algorithms, pseudocode, flowcharts, test cases, source code, program output, and terminal instructions.

Part 1 - Time Converter: Algorithm

HDOC DAY-10 ACTIVITY REPORT: BATCHES 15 & 16

PART 1: TIME CONVERTER PROGRAM

1.1 Algorithm

1. **Start**
2. **Input:** Read total seconds (`total_seconds`).
3. **Calculate Days:** `days = total_seconds / 86400`
4. **Calculate Remaining Seconds:** `remaining_seconds = total_seconds % 86400`
5. **Calculate Hours:** `hours = remaining_seconds / 3600`
6. **Update Remaining Seconds:** `remaining_seconds = remaining_seconds % 3600`
7. **Calculate Minutes:** `minutes = remaining_seconds / 60`
8. **Calculate Seconds:** `seconds = remaining_seconds % 60`
9. **Output:** Display days, hours, minutes, and seconds.
10. **Stop**

Part 1 - Pseudocode

1.2 Pseudocode

text

BEGIN

```
NUMBER total_seconds, days, hours, minutes, seconds
OUTPUT "Enter time in seconds: "
INPUT total_seconds
```

```
days = total_seconds / 86400
hours = (total_seconds % 86400) / 3600
minutes = ((total_seconds % 86400) % 3600) / 60
seconds = ((total_seconds % 86400) % 3600) % 60
```

Part 1 - Flowchart and Test Cases

```
    OUTPUT days, "Days,", hours, "Hours,", minutes,  
    "Minutes,", seconds, "Seconds"  
END
```

Use code with caution.

1.3 Flowchart Text Representation

text

```
[ Start ]  
  |  
  ▼  
[ Input total_seconds ]  
  |  
  ▼  
[ days = total_seconds / 86400 ]  
[ rem_seconds = total_seconds % 86400 ]  
  |  
  ▼  
[ hours = rem_seconds / 3600 ]  
[ rem_seconds = rem_seconds % 3600 ]  
  |  
  ▼  
[ minutes = rem_seconds / 60 ]  
[ seconds = rem_seconds % 60 ]  
  |  
  ▼  
[ Print Days, Hours, Minutes, Seconds ]  
  |  
  ▼  
[ Stop ]
```

Use code with caution.

1.4 Test Case Table

Test Case ID	Input (Seconds)	Expected Output	Status
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Part 1 - Test Cases, Source Code and Output

TC1_01	0	0 Days, 0 Hours, 0 Minutes, 0	Pass
TC1_02	45	0 Days, 0 Hours, 0 Minutes, 45 Seconds	Pass
TC1_03	3750	0 Days, 1 Hours, 2 Minutes, 30 Seconds	Pass
TC1_04	90061	1 Days, 1 Hours, 1 Minutes, 1	Pass

1.5 Source Code (time_converter.c)

C

```

LM F100 0.09 File: day10.c Modified
#include <stdio.h>

int main() {
    long long total_seconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    if (scanf("%lld", &total_seconds) != 1 || total_seconds < 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 1;
    }

    days = total_seconds / 86400;
    int rem_seconds = total_seconds % 86400;

    hours = rem_seconds / 3600;
    rem_seconds = rem_seconds % 3600;

    minutes = rem_seconds / 60;
    seconds = rem_seconds % 60;

    printf("Total seconds = %d Days, %d Hours, %d Minutes, %d Seconds\n",
        total_seconds, days, hours, minutes, seconds);

    return 0;
}

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Sep 3 16:12:05 on console
Wed Sep  2 23:43:45 IST 2026
ms-MacBook-Air ~ % nano day10.c
ms-MacBook-Air ~ % clang day10.c -o day10
ms-MacBook-Air ~ % ./day10
seconds: 12
ms-MacBook-Air ~ % ./day10
days: 0 hours: 0 minutes: 12 seconds: 12
ms-MacBook-Air ~ % ./day10
seconds: 36
ms-MacBook-Air ~ % ./day10
days: 0 hours: 0 minutes: 36 seconds: 36
ms-MacBook-Air ~ % ./day10
seconds: 7
ms-MacBook-Air ~ % ./day10
days: 0 hours: 0 minutes: 7 seconds: 7
ms-MacBook-Air ~ % ./day10
seconds: 99
ms-MacBook-Air ~ % ./day10
days: 0 hours: 1 minutes: 39 seconds: 39
ms-MacBook-Air ~ % ./day10
days: 0 hours: 1 minutes: 39 seconds: 39
ms-MacBook-Air ~ % ./day10
seconds: 45
ms-MacBook-Air ~ % ./day10
days: 0 hours: 0 minutes: 45 seconds: 45
ms-MacBook-Air ~ % ./day10
seconds: 3750
ms-MacBook-Air ~ % ./day10
days: 1 hours: 2 minutes: 30 seconds: 30
ms-MacBook-Air ~ % ./day10
seconds: 90061
ms-MacBook-Air ~ % ./day10
days: 1 hours: 1 minutes: 1 seconds: 1
ms-MacBook-Air ~ %

```

Part 2 - Leap Year Checker: Algorithm and Pseudocode

PART 2: LEAP YEAR CHECKER (COMPACT & LOGICAL OPTIMIZED)

2.1 Algorithm

1. **Start**
2. **Input:** Read year.
3. **Condition Check:** If (year % 4 == 0 AND year % 100 != 0) OR (year % 400 == 0), flag as Leap Year.
4. **Else:** Flag as Not a Leap Year.
5. **Output:** Print validation result.
6. **Stop**

2.2 Pseudocode

text

BEGIN

 NUMBER year

 OUTPUT "Enter a year: "

 INPUT year

 IF ((year % 4 == 0 AND year % 100 != 0) OR (year % 400 == 0)) THEN

 OUTPUT year, " is a leap year."

 ELSE

 OUTPUT year, " is not a leap year."

 ENDIF

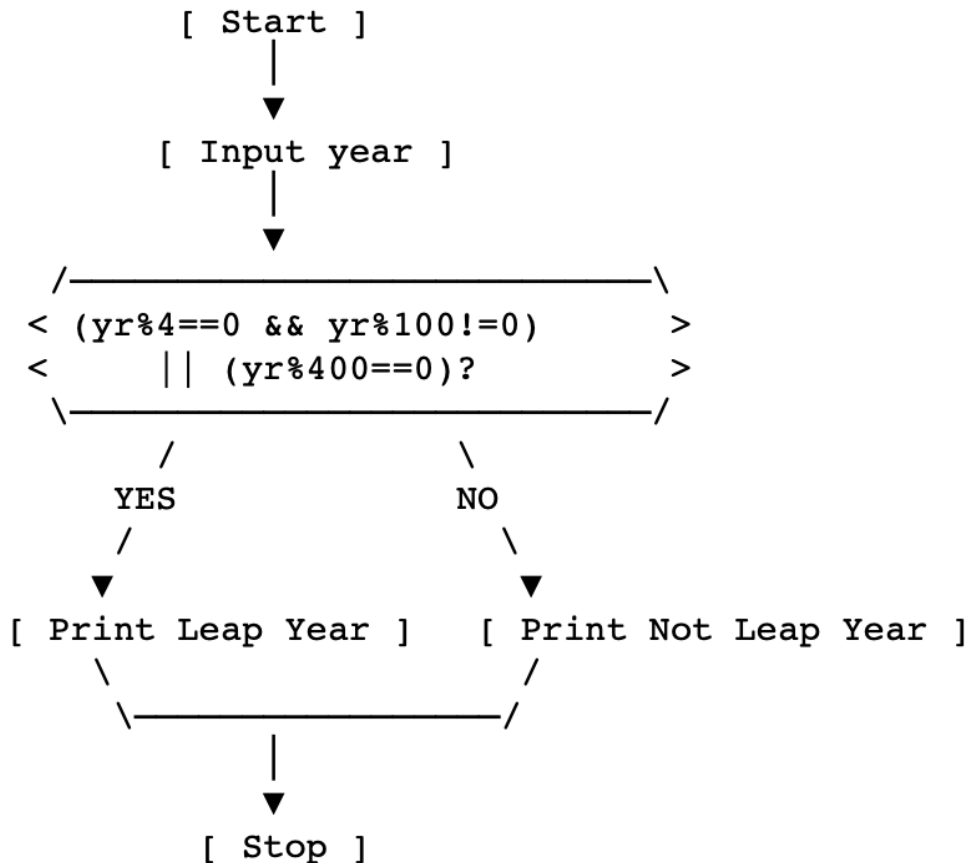
END

Use code with caution.

Part 2 - Flowchart and Test Cases

2.3 Flowchart Text Representation

text



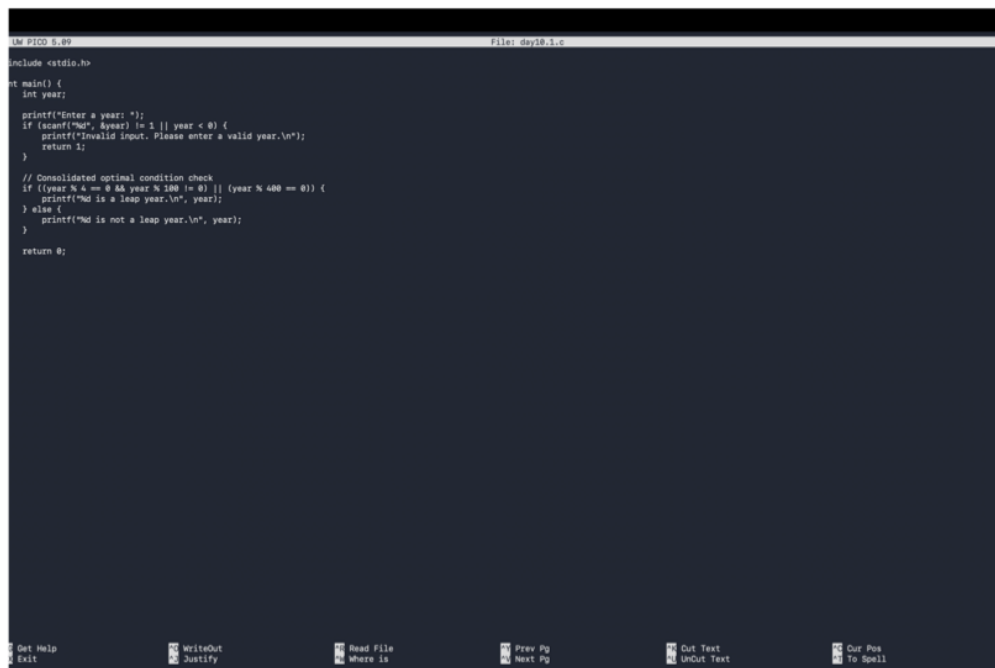
Use code with caution.

2.4 Test Case Table

Test Case ID	Input (Year)	Expected Output	Status
TC2_01	2024	2024 is a leap year.	Pass
TC2_02	1900	1900 is not a leap year.	Pass
TC2_03	2000	2000 is a leap year.	Pass
TC2_04	2023	2023 is not a leap year.	Pass

2.5 Source Code (`leap_year.c`)

Part 2 - Source Code



```
LM PICO 3.00 File: day10.1.c
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    if (scanf("%d", &year) != 1 || year < 0) {
        printf("Invalid input. Please enter a valid year.\n");
        return 1;
    }

    // Consolidated optimal condition check
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```

Caption

Part 2 - Program Execution Output

```
[gurveer@gurveer MacBook Air ~ % clang day10.1.c -o day10.1
[gurveer@gurveer MacBook Air ~ % ./day10.1
Enter a year: 2024
2024 is a leap year.
[gurveer@gurveer MacBook Air ~ % ./day10.1
Enter a year: 1998
1998 is not a leap year.
[gurveer@gurveer MacBook Air ~ % ./day10.1
Enter a year: 2000
2000 is a leap year.
[gurveer@gurveer MacBook Air ~ % ./day10.1
Enter a year: 2023
2023 is not a leap year.
[gurveer@gurveer MacBook Air ~ % ]
```

Part 3 - Compilation and Execution Instructions

PART 3: COMPILATION & EXECUTION INSTRUCTIONS

To complete the execution step required for Day-10, students can run the following terminal sequences:

bash

```
# Compile the programs
gcc time_converter.c -o time_converter
gcc leap_year.c -o leap_year
```

```
# Execute the programs
./time_converter
./leap_year
```

STUDENT DETAILS

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